

Name: _____

Exam Style Questions

3D Trigonometry



Equipment needed: Pen, Calculator, Ruler, Pencil

Guidance

1. Read each question carefully before you begin answering it.
2. Check your answers seem right.
3. Always show your workings

Video Tutorial

www.corbettmaths.com/contents

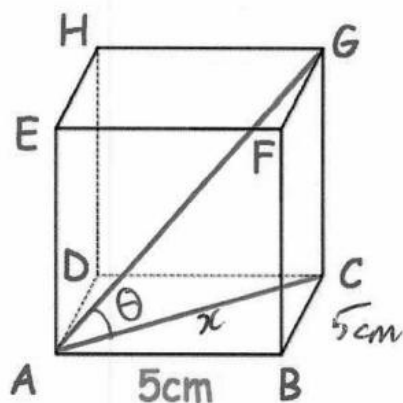
Video 332



Answers and Video Solutions



1. Shown is a cube with side length 5cm.



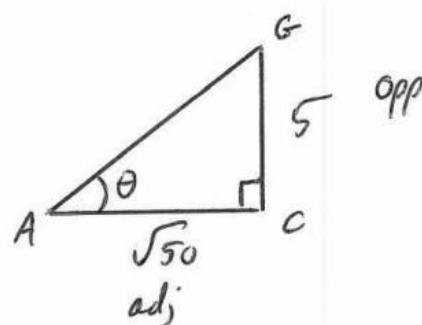
Calculate angle CAG.

$$x^2 = 5^2 + 5^2$$

$$x^2 = 25 + 25$$

$$x^2 = 50$$

$$x = \sqrt{50}$$



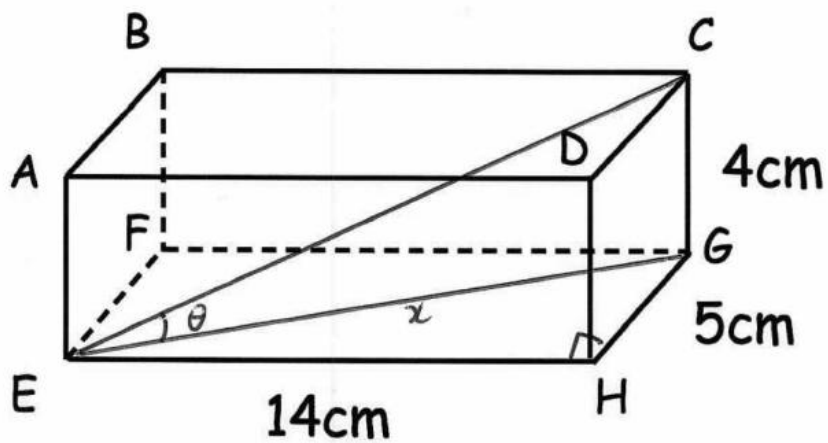
$$\tan \theta = \frac{5}{\sqrt{50}}$$

$$\theta = 35.264 \dots$$

$$\underline{35.264}^{\circ}$$

(4)

2. Shown below is a cuboid.

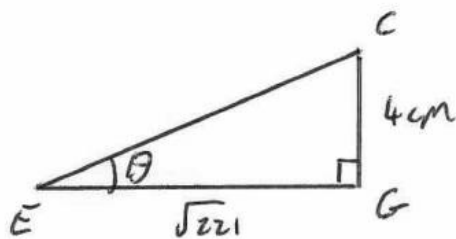


Calculate the size of angle CEG

$$x^2 = 5^2 + 14^2$$

$$x^2 = 221$$

$$x = \sqrt{221}$$



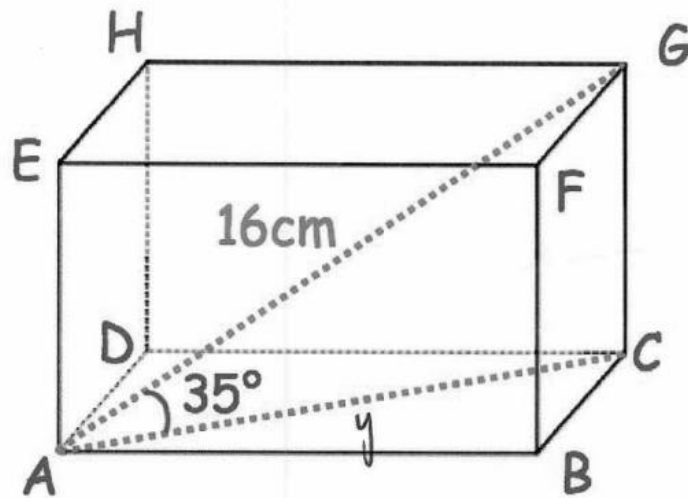
$$\tan \theta = \frac{4}{\sqrt{221}}$$

$$\theta = 15.0598\dots$$

$$15.06$$

°
(4)

3. Shown below is a cuboid.

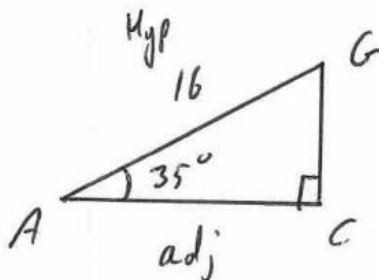


Length AG = 16cm

Angle CAG is 35°

$$EG = AC$$

Work out the length of EG.



$$\cos(35) = \frac{y}{16}$$

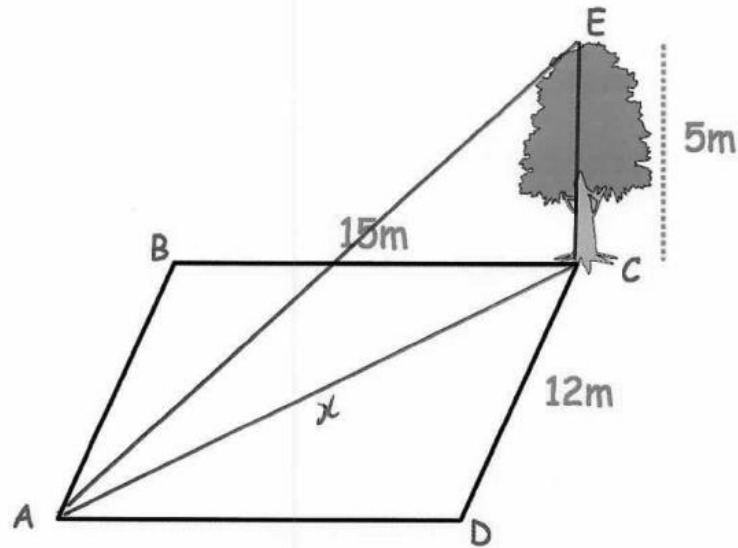
$$y = 16 \times \cos(35)$$

$$y = 13.1064..$$

$$\underline{\underline{13.106}} \text{ cm}$$

(3)

4. A tree is located in the corner of a rectangular field.



The field is 15 metres long and 12 metres wide.
The tree is 5 metres tall.

Calculate angle CAE.

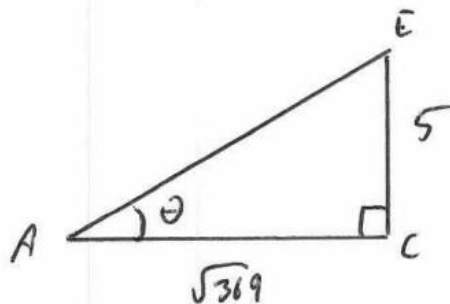
$$x^2 = 12^2 + 15^2$$

$$x^2 = 369$$

$$x = \sqrt{369}$$

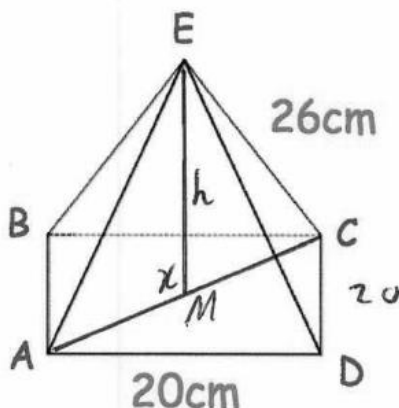
$$\tan \theta = \frac{5}{\sqrt{369}}$$

$$\theta = 14.5897\dots$$



$$\frac{14.59}{(4)}$$

5. Shown below is a square based pyramid.
The apex E is directly over the centre of the base.



$$AD = 20\text{cm}$$

$$CE = 26\text{cm}$$

- (a) Work out the length of AC

$$x^2 = 20^2 + 20^2$$

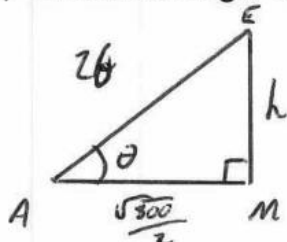
$$x^2 = 800$$

$$x = \sqrt{800}$$

$$28.2842\dots\text{cm}$$

(2)

- (b) Calculate angle CAE



$$\cos \theta = \frac{\left(\frac{\sqrt{800}}{2}\right)}{26}$$

$$\theta = 57.0485\dots$$

$$57.0485\dots^\circ$$

(2)

- (c) Work out the height of the pyramid

$$h^2 + \left(\frac{\sqrt{800}}{2}\right)^2 = 26^2$$

$$h^2 = 476$$

$$h = \sqrt{476}$$

$$21.8174\dots\text{cm}$$

(2)

- (d) Calculate the volume of the pyramid

$$V = \frac{1}{3} A h$$

$$A = 20 \times 20 = 400\text{cm}^2$$

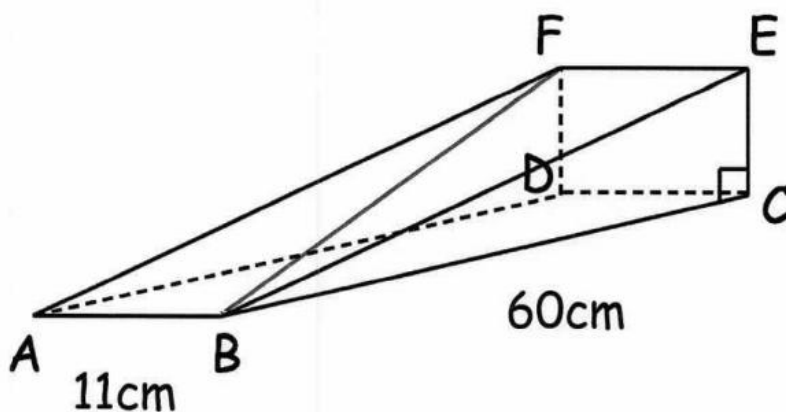
$$= \frac{1}{3} \times 400 \times \sqrt{476}$$

$$= 2908.9898\dots$$

$$2908.99\text{cm}^3$$

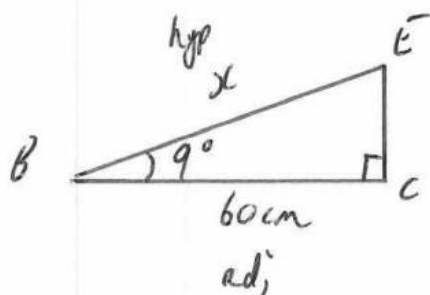
(2)

6. A triangular prism is shown below.



Angle CBE = 9°

- (a) Calculate the length of BE.



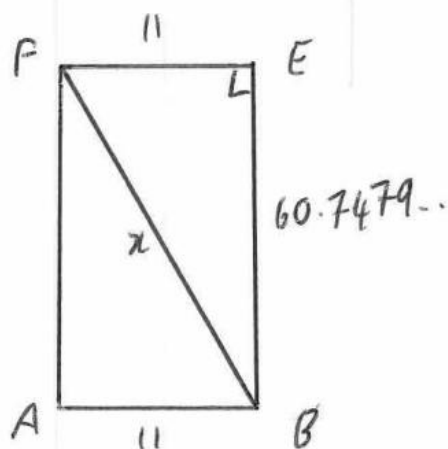
$$\cos(9) = \frac{60}{x}$$

$$x = \frac{60}{\cos(9)}$$

$$x = 60.7479\dots$$

$$\underline{60.7479} \text{ cm} \quad (2)$$

- (b) Calculate the length of BF.



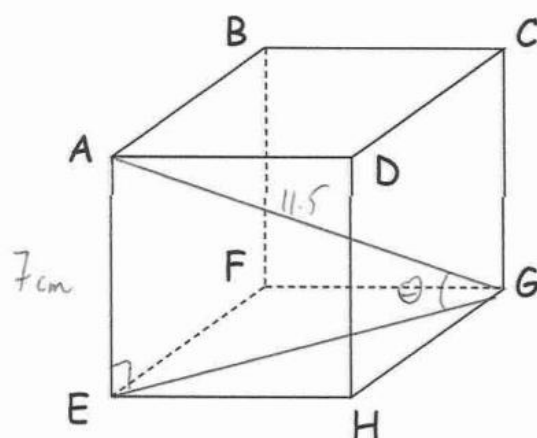
$$x^2 = 11^2 + 60.7479\dots^2$$

$$x^2 = 3811.308271$$

$$x = 61.73579409$$

$$\underline{61.736} \text{ cm} \quad (2)$$

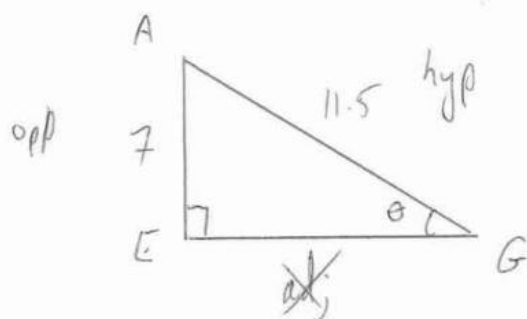
7. Shown below is a cuboid.



$$AE = 7\text{cm}$$

$$AG = 11.5\text{cm}$$

Work out the size of the angle between AG and plane EFGH.



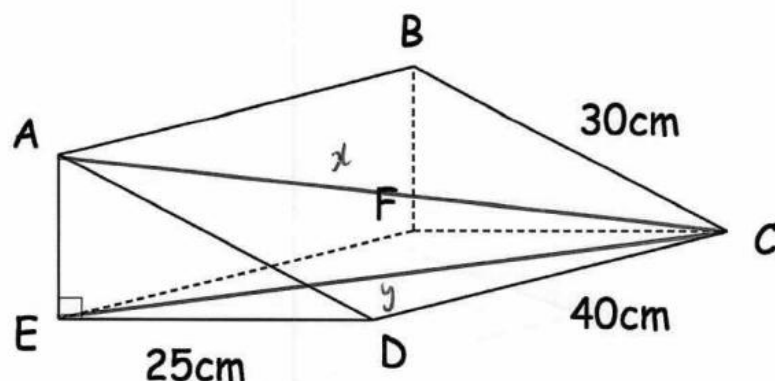
$$\sin \theta = \frac{\text{opp}}{\text{hyp}}$$

$$\sin \theta = \frac{7}{11.5}$$

$$= 37.495 \dots^\circ$$

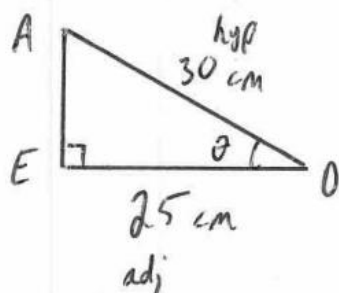
$$\frac{37.495^\circ}{(2)}$$

8. Here is a triangular prism.



Which angle is larger, angle ADE or angle ACE?

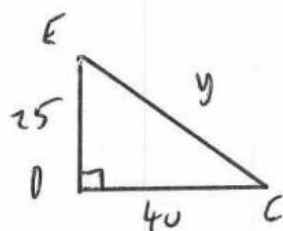
You must show your working.



$$\cos \theta = \frac{25}{30}$$

$$\theta = 33.5573\dots^\circ$$

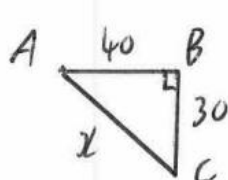
$$\angle ADE = 33.5573^\circ$$



$$y^2 = 25^2 + 40^2$$

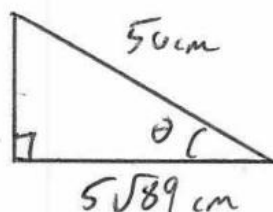
$$y^2 = 2225$$

$$y = 5\sqrt{89}$$



$$x^2 = 30^2 + 40^2$$

$$x = 50$$



$$\cos \theta = \frac{5\sqrt{89}}{50}$$

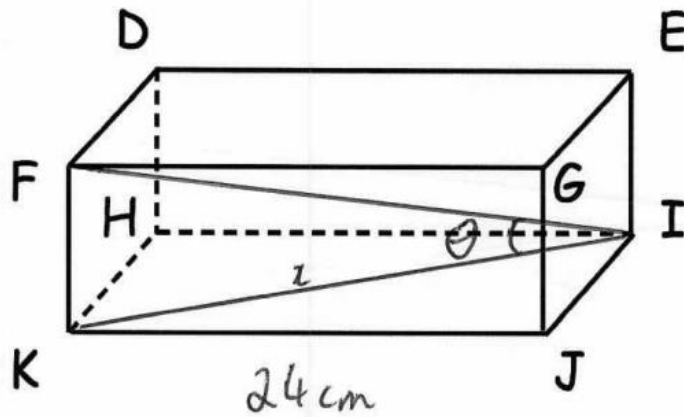
$$\theta = 19.3697\dots$$

$$\angle ACE = 19.3697^\circ$$

(4)

$\angle ADE$ is larger than $\angle ACE$

9. Shown below is a cuboid.



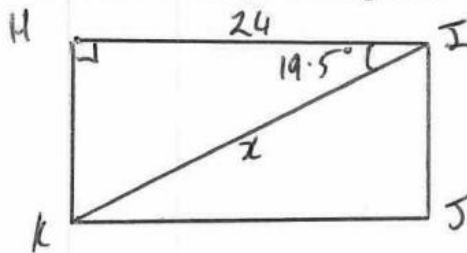
$$KJ = 24\text{cm}$$

$$KG = 25\text{cm}$$

$$\text{Angle } HIK = 19.5^\circ$$

Calculate the size of angle KIF

base

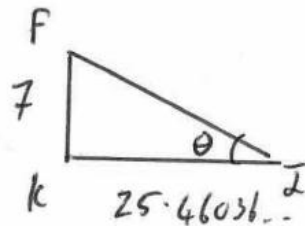
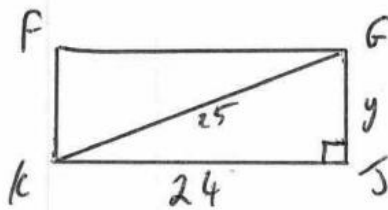


$$\cos(19.5) = \frac{24}{x}$$

$$x = \frac{24}{\cos(19.5)}$$

$$x = 25.460368\dots$$

front



$$\tan \theta = \frac{7}{25.46036\dots}$$

$$\theta = 15.3729$$

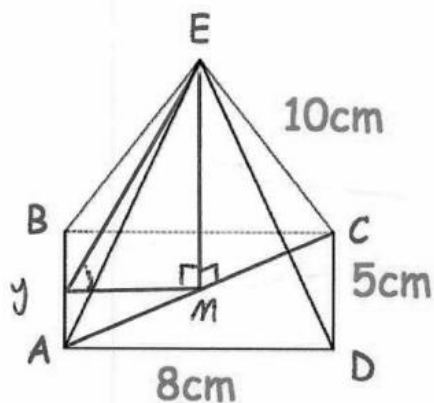
$$24^2 + y^2 = 25^2$$

$$y^2 = 49$$

$$y = 7$$

$$\frac{15.3729}{(4)}$$

10. Shown below is a rectangular based pyramid.
The apex E is directly over the centre of the base.



AD = 8cm
CD = 5cm
CE = 10cm

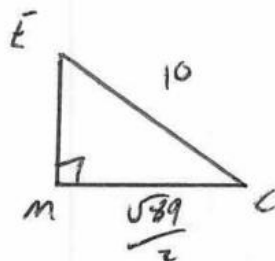
- (a) Calculate the height of the pyramid

$$AC^2 = 5^2 + 8^2$$

$$AC^2 = 89$$

$$AC = \sqrt{89}$$

$$MC = \frac{\sqrt{89}}{2}$$



$$EC^2 = ME^2 + MC^2$$

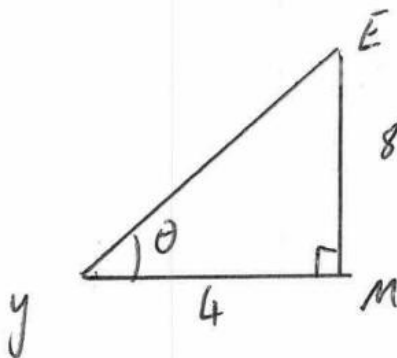
$$100 = ME^2 + \frac{89}{4}$$

$$ME^2 = \frac{311}{4}$$

$$ME = 8.817596...$$

$$\underline{\underline{8.818}} \text{ cm to 3dp} \quad (4)$$

- (b) Calculate angle between the face ABE and the base ABCD



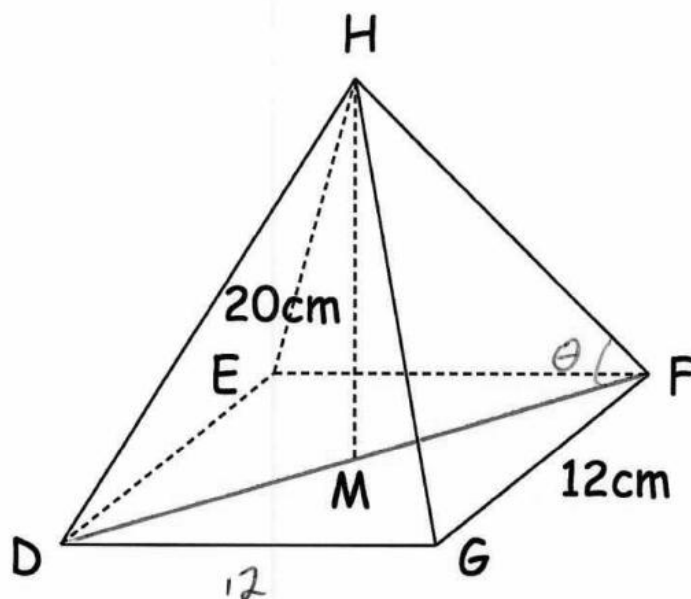
$$8.817596...$$

$$\tan \theta = \frac{8.817...}{4}$$

$$\theta = 65.5991...$$

$$\underline{\underline{65.6}}^\circ \quad (3)$$

11. Shown below is a square based pyramid.
 The side length of the square is 12cm.
 The perpendicular height of the pyramid is 20cm.



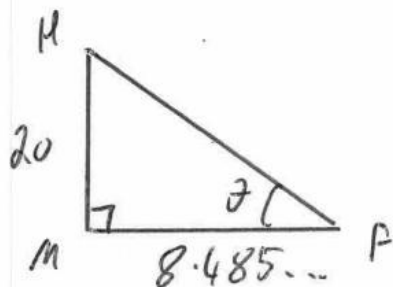
Work out the size of angle HFD.

$$DF^2 = 12^2 + 12^2$$

$$DF^2 = 288$$

$$DF = 12\sqrt{2} \quad (16.9705\dots)$$

$$FM = 6\sqrt{2} \quad (8.485\dots)$$

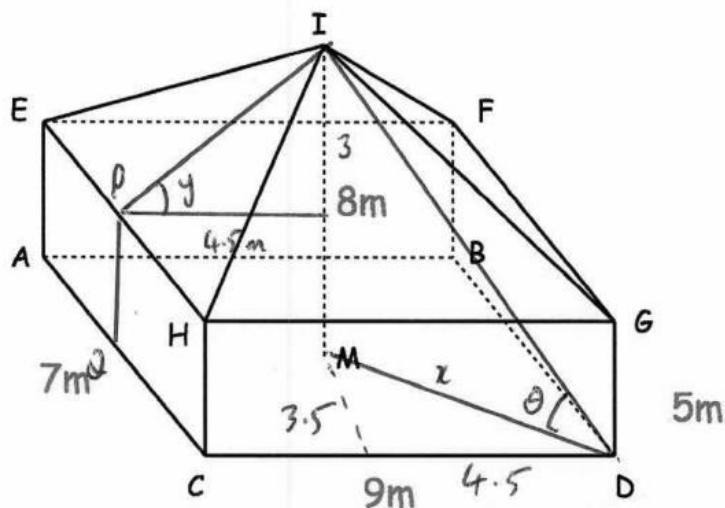


$$\tan \theta = \frac{20}{8.485\dots}$$

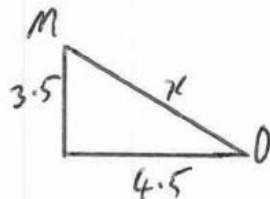
$$\theta = 67.010\dots$$

$$\underline{\underline{67.01^\circ}} \quad (4)$$

12. The diagram shows a cuboid and a pyramid.
 The apex of the pyramid, I, is directly above the centre, M, of ABDC.



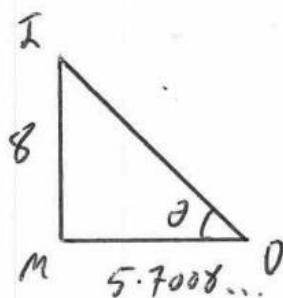
- (a) Calculate the angle between the line DI and the plane ABDC



$$x^2 = 3.5^2 + 4.5^2$$

$$x^2 = 32.5$$

$$x = 5.7008...$$



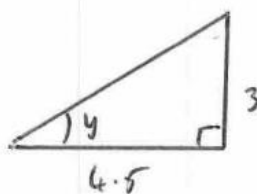
$$\tan \theta = \frac{8}{5.7008...}$$

$$\theta = 54.5259...$$

$$\underline{54.5259...}$$

(4)

- (b) Calculate the angle between planes EHI and ACHE



$$\tan y = \frac{3}{4.5}$$

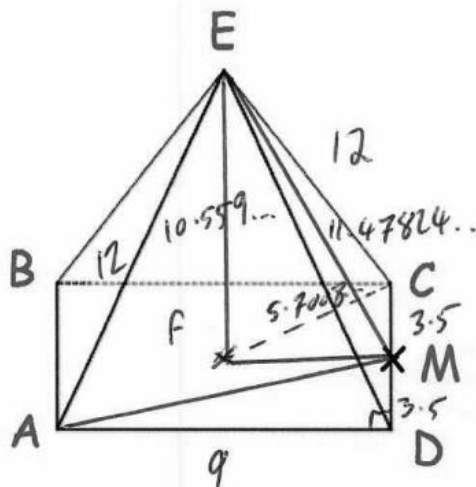
$$y = 33.690...$$

$$\angle QPI = 90 + 33.69...$$

$$\underline{123.69...}$$

(4)

13. Shown below is a rectangular-based pyramid.
The apex E is directly over the base of the pyramid.



$$AD = 9\text{cm}$$

$$CD = 7\text{cm}$$

$$CE = 12\text{cm}$$

M is the midpoint of the line CD.

Work out the size of angle AME

$$CF^2 = CM^2 + MF^2$$

$$CF^2 = 3.5^2 + 4.5^2$$

$$CF^2 = 32.5$$

$$CF = 5.700877125$$

$$EF^2 = CF^2 + EF^2$$

$$12^2 = 5.700877125^2 + h^2$$

$$h^2 = \frac{223}{2}$$

$$h = 10.55935604$$

$$EF = 10.55935604$$

$$EM^2 = 4.5^2 + 10.55935604^2$$

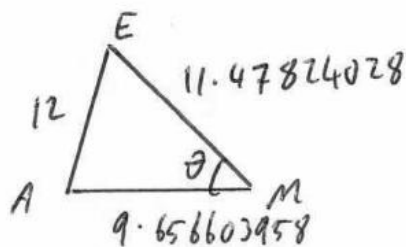
$$EM^2 = \frac{527}{4}$$

$$EM = 11.47824028$$

$$AM^2 = 9^2 + 3.5^2$$

$$AM^2 = \frac{373}{4}$$

$$AM = 9.6566...$$



$$68.568$$

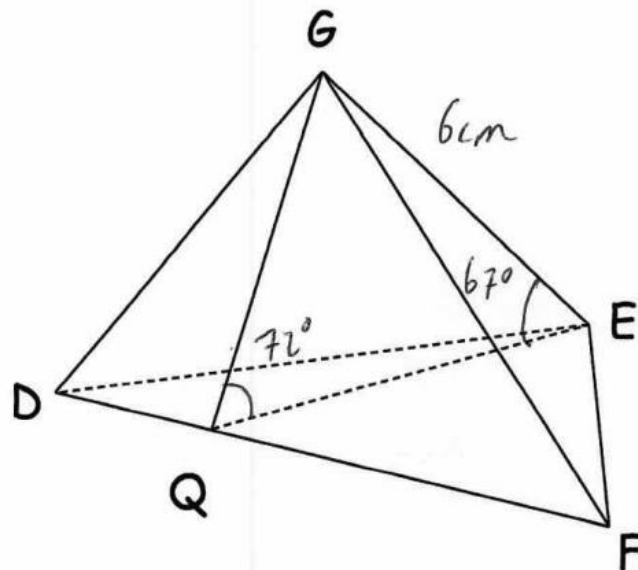
(4)

$$\cos \theta = \frac{(11.47...)^2 + (9.656...)^2 - 12^2}{2 \times (11.47...) \times (9.656...)}$$

$$\cos \theta = 0.365389...$$

$$\theta = 68.568...$$

14. DEFG is a triangular based pyramid.



Q is a point on DF.

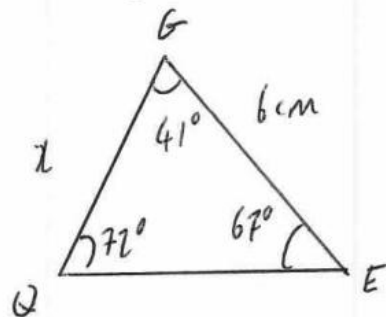
$$GE = 6\text{cm}$$

$$\text{Angle } GEQ = 67^\circ$$

$$\text{Angle } GQE = 72^\circ$$

Work out the length of GQ.

$$180 - 72 - 67 = 41^\circ$$



$$\frac{x}{\sin 67} = \frac{6}{\sin 72}$$

$$x = 5.80725\dots$$

$$5.8072\dots$$

$$\dots\dots\dots\text{cm}$$

(3)